

			Total Pages: 3
Draft Scheme of Valuation/Answer Key (Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY FOURTH SEMESTER B.TECH DEGREE EXAMINATION, MAY 2021			
Course Code:CST 206			
Course Name: OPERATING SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		<i>(Answer all questions; each question carries 3 marks)</i>	Marks
1		Three Methods - (1) using registers (2) parameters are stored as a block in memory and address of that block is passed (3) using stack 1 mark each	3
2		Any three points	3
3		Difference- 2 marks, Application- 1 mark	3
4		Reason with Explanation	3
5		Mutual Exclusion, hold and wait, circular wait and non-pre-emption	3
6		Wait and signal operations-each carry 1.5 marks.	3
7		Swapping - Explanation.	3
8		Virtual memory concept – 2marks example : demand paging - 1mark	3
9		Comparison – Sequential access vs direct access	3
10		Physical formatting , partition, logical formatting (1+1+1)	3
PART B			
<i>(Answer one full question from each module, each question carries 14 marks)</i>			
Module -1			
11	a)	Explanation with difference of each carry 3 marks (Multiprogramming systems, Multitasking systems, Multiprocessor systems.)	9
	b)	The long-term scheduler directly affects the system performance. Explain how	5
12	a)	Explanation of any six functions	6
	b)	Operating system structures- Layered – 4 marks Microkernel – 4 marks	8
Module -2			

13	a)	Explanation	8																														
	b)	7 child process will be created – 1 marks HELLO will be printed 4 times - 1.5 mark WELCOME will be printed 8 times. - 1.5 mark Justification of answer – 2 marks	6																														
14	a)	Drawing Gantt chart - 4 marks (1+1+1+1) RR waiting time=10 (2 marks) FCFS waiting time=8.2 (2 marks) Priority waiting time=10 (2 marks) SJF waiting time=5.4 (2 marks)	12																														
	b)	In this case, the sender must block until the recipient receives the message.	2																														
Module -3																																	
15	a)	Total instances (A, B, C, D) = (13, 8, 9, 8) – 1 mark Need matrix (2 marks) <table border="1"><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>P1</td><td>2</td><td>2</td><td>3</td><td>0</td></tr><tr><td>P2</td><td>3</td><td>2</td><td>0</td><td>0</td></tr><tr><td>P3</td><td>0</td><td>3</td><td>2</td><td>3</td></tr><tr><td>P4</td><td>2</td><td>5</td><td>0</td><td>7</td></tr><tr><td>P5</td><td>2</td><td>0</td><td>0</td><td>1</td></tr></table> The system is in safe state and the safe sequence is -P5, P2, P3, P1, P4 - 1mark Steps with proper explanation. (4 marks)		A	B	C	D	P1	2	2	3	0	P2	3	2	0	0	P3	0	3	2	3	P4	2	5	0	7	P5	2	0	0	1	8
	A	B	C	D																													
P1	2	2	3	0																													
P2	3	2	0	0																													
P3	0	3	2	3																													
P4	2	5	0	7																													
P5	2	0	0	1																													
	b)	Critical section problem - 1 mark Three requirements - 2 marks Solution to a 2 process critical section problem(Petersons solution) – 3 marks	6																														
16	a)	Description of Bounded-buffer problem (2 marks) + solution for the same using Semaphores (3 marks) + structure of producer and consumer processes. (3 marks)	8																														
	b)	Reason (2 marks) + Resource allocation graph (i) deadlock (ii) with cycle but no deadlock (2+2 marks)	6																														

Module -4			
17	a)	Diagram – 2 marks . Explanation – 6 marks	8
	b)	Diagram – 2 marks . Steps – 4 marks	6
18	a)	FCFS-16 page faults, LRU-15 page faults, Optimal-11 page faults, Each carry three marks, final output-1 mark and showing the page faults-2 marks	9
	b)	(i) 1850 (ii) 800 (iii) 2670 (iv) 125 (v) 3830 - 1 marks each	5
Module -5			
19	a)	FCFS-Total seek time=642 SSTF-Total seek time=208 SCAN,(50->82->140->170->190->199->43->24->16) total seek time=324, if initial movement is towards cylinder number 199 SCAN (50->43->24->16->0->82->140->170->190) total seek time = 220, if initial movement is towards cylinder number 0. Explaining each carry 2 mark and output carry 1 mark.	9
	b)	Explanation with diagram.	5
20	a)	Explanation of single-level, two-level, tree, acyclic graph and general graph (each carry 2 marks)	10
	b)	Access Controls list, (owner, group, universe) – 4 marks	4
